

Lecture 3: Limit of Sequence

Mathematical Analysis

1 Convergent Sequences

Definition 1.1 (Convergent Sequence). [def:convergent-sequence] A sequence $a : \mathbb{N}_1 \rightarrow F$, where F is a ordered field, is said to be *convergent* if

$$\exists \ell \in F, \forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1, n \geq N, d(a_n, \ell) < \varepsilon.$$

Remark 1.1 (The ε - N Challenge). [rem:epsilon-n-challenge] A value ℓ passes the ε - N challenge if for any proposed $\varepsilon > 0$, there exists an index $N \in \mathbb{N}_1$ such that for all $n \in \mathbb{N}_1$ with $n \geq N$, $d(a_n, \ell) < \varepsilon$.

Remark 1.2. In the The ε - N Challenge, the value of N may depend on the choice of ε . And the value N may be not unique (ie, $N' > N$ also works)

Theorem 1.1 (Uniqueness of Limit). [thm:limit-uniqueness] Let F be an ordered field, $a : \mathbb{N}_1 \rightarrow F$, and $\ell_1, \ell_2 \in F$. Then

$$\begin{aligned} &(\forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1, n \geq N, d(a_n, \ell_1) < \varepsilon) \\ &\quad \wedge \\ &(\forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1, n \geq N, d(a_n, \ell_2) < \varepsilon) \\ &\implies \ell_1 = \ell_2. \end{aligned}$$

Proof. Assume for contradiction that $\ell_1 \neq \ell_2$. Set

$$\varepsilon = \frac{d(\ell_1, \ell_2)}{2}.$$

Since $\ell_1 \neq \ell_2$, we have $\varepsilon > 0$.

By the hypothesis for ℓ_1 , there exists $N_1 \in \mathbb{N}_1$ such that for all $n \geq N_1$, $d(a_n, \ell_1) < \varepsilon$. Similarly, there exists $N_2 \in \mathbb{N}_1$ such that for all $n \geq N_2$, $d(a_n, \ell_2) < \varepsilon$.

Let $N = \max(N_1, N_2)$. For any $n \geq N$, both hypotheses hold:

$$d(a_n, \ell_1) < \varepsilon \quad \text{and} \quad d(a_n, \ell_2) < \varepsilon.$$

By the triangle inequality for distance applied via the intermediate point a_n ,

$$d(\ell_1, \ell_2) \leq d(\ell_1, a_n) + d(a_n, \ell_2) = d(a_n, \ell_1) + d(a_n, \ell_2),$$

where the last equality uses symmetry of distance. Combining,

$$d(\ell_1, \ell_2) \leq d(a_n, \ell_1) + d(a_n, \ell_2) < 2\varepsilon = d(\ell_1, \ell_2),$$

a contradiction. Hence $\ell_1 = \ell_2$. □

Remark 1.3 (Compact Form Using $\exists!$). [rem:limit-uniqueness-compact] Combined with the existence from the definition of convergent sequence, this theorem states: if a is convergent, then there exists a unique such ℓ :

$$\exists! \ell \in F, \forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1, n \geq N, d(a_n, \ell) < \varepsilon.$$

Here $\exists!$ is shorthand for “existence and uniqueness”; it should be formally introduced in lec01 as a FOL convention before being used.

2 Convergence Challenge Functions

Definition 2.1 (Convergence Challenge Function). [def:convergence-challenge-function] Let F be an ordered field. A function $g : F \rightarrow F$ is a *convergence challenge function* if it satisfies two conditions:

- *Positivity on positives*: $\forall \gamma \in F, \gamma > 0 \implies g(\gamma) > 0$.
- *Arbitrarily small values*: $\forall \delta \in F, \delta > 0, \exists \gamma \in F, \gamma > 0 \wedge g(\gamma) < \delta$.

Remark 2.1 (Intuition: Covering Arbitrarily Small Positives). [rem:challenge-intuition]

Theorem 2.1 (Convergence Challenge Replacement). [thm:convergence-challenge-replacement] Let F be an ordered field, $a : \mathbb{N}_1 \rightarrow F$, and let g be a convergence challenge function. Then

$$(\exists \ell \in F, \forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1, n \geq N, d(a_n, \ell) < \varepsilon)$$

$$\iff$$

$$(\exists \ell \in F, \forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1, n \geq N, d(a_n, \ell) < g(\varepsilon)).$$

Proof. Fix the witness $\ell \in F$ in either direction.

($\implies$) For $\varepsilon > 0$, the positivity-on-positives condition gives $g(\varepsilon) > 0$. Apply the definition of convergent sequence at the challenge $g(\varepsilon)$.

($\impliedby$) For $\varepsilon > 0$, the challenge-function property of g gives $\gamma > 0$ with $g(\gamma) < \varepsilon$. The hypothesis at γ gives N such that

$$d(a_n, \ell) < g(\gamma) < \varepsilon \quad \text{for all } n \geq N.$$

□

Lemma 2.1 (Linear Scaling). [lem:linear-is-challenge] For any $c \in F$ with $c > 0$, the function $g(\gamma) = c\gamma$ is a convergence challenge function.

Proof. We verify both conditions.

Positivity on positives. For any $\gamma \in F$ with $\gamma > 0$, both $c > 0$ and $\gamma > 0$, so $g(\gamma) = c\gamma > 0$.

Arbitrarily small values. Fix any $\delta \in F$ with $\delta > 0$. We seek $\gamma \in F$ with $\gamma > 0$ and $g(\gamma) < \delta$.

Take $\gamma := \delta/(c+1)$, which is derived in remark 2.2. Since $\delta > 0$ and $c+1 > 0$ (because $c > 0$), we have $\gamma > 0$.

Now $g(\gamma) = c\gamma = c\delta/(c+1)$. Since $c < c+1$, we have $c/(c+1) < 1$, hence $c\delta/(c+1) < \delta$. □

Remark 2.2 (Deriving the Witness for Linear Scaling). [rem:linear-scaling-witness-derivation] To find the witness γ used in the proof of lemma 2.1, we start from the target inequality $g(\gamma) < \delta$ and manipulate it to isolate γ :

$$g(\gamma) < \delta \iff c\gamma < \delta \iff \gamma < \frac{\delta}{c}.$$

Any γ in the open interval $(0, \frac{\delta}{c})$ will therefore satisfy $g(\gamma) < \delta$. To pick a concrete such γ , we use the fact that $c+1 > c$, which gives

$$\frac{\delta}{c+1} < \frac{\delta}{c}.$$

Hence we may take $\gamma := \delta/(c+1)$ as the witness.

Remark 2.3 (Why $c \leq 0$ is Excluded). [rem:linear-scaling-cnonpos] If $c \leq 0$, then for any $\gamma \in F$ with $\gamma > 0$, $g(\gamma) = c\gamma \leq 0$. This violates the positivity-on-positives condition, so g is not a convergence challenge function.

Lemma 2.2 (Power). [lem:power-is-challenge] For any positive integer k , the function $g(\gamma) = \gamma^k$ is a convergence challenge function.

Proof. Both conditions follow by induction on k . Positivity: $\gamma^k > 0$ for $\gamma > 0$. For arbitrarily small values, take $\gamma := \min(\delta/2, 1/2)$, so $\gamma > 0$ and $\gamma \leq 1$; induction gives $\gamma^k \leq \gamma \leq \delta/2 < \delta$. $\square$

Lemma 2.3 (Summation). [lem:sum-is-challenge] Let f, g be convergence challenge functions, and suppose both are monotone increasing (i.e. $x \leq y \implies f(x) \leq f(y)$, and similarly for g). Then $f + g$, defined by $(f + g)(\gamma) := f(\gamma) + g(\gamma)$, is a convergence challenge function.

Proof. Positivity is immediate from $f(\gamma), g(\gamma) > 0$. For $\delta > 0$, pick $\gamma_1, \gamma_2 > 0$ with $f(\gamma_1), g(\gamma_2) < \delta/2$; set $\gamma := \min(\gamma_1, \gamma_2)$. Monotonicity gives $(f + g)(\gamma) \leq f(\gamma_1) + g(\gamma_2) < \delta$. $\square$

Lemma 2.4 (Composition). [lem:challenge-composition] Let f, g be convergence challenge functions, and suppose f is monotone increasing (i.e. $x \leq y \implies f(x) \leq f(y)$). Then $f \circ g$ is a convergence challenge function.

Proof. Positivity: $f(g(\gamma)) > 0$ since $g(\gamma) > 0$ and f preserves positivity. For $\delta > 0$, pick $\eta > 0$ with $f(\eta) < \delta$, then $\gamma > 0$ with $g(\gamma) < \eta$. Monotonicity of f gives $(f \circ g)(\gamma) \leq f(\eta) < \delta$. $\square$

Theorem 2.2 (Closed Form of Convergence). [thm:convergence-closed-form] Let F be an ordered field, let $a : \mathbb{N}_1 \rightarrow F$. Then

$$\begin{aligned} & (\exists \ell \in F, \forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1, n \geq N, d(a_n, \ell) < \varepsilon) \\ & \iff \\ & (\exists \ell \in F, \forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1, n \geq N, d(a_n, \ell) \leq \varepsilon). \end{aligned}$$

Proof. $(\implies)$ If $d(a_n, \ell) < \varepsilon$, then $d(a_n, \ell) \leq \varepsilon$.

$(\impliedby)$ By lemma 2.1 with $c = 2$, $g(\varepsilon) = 2\varepsilon$ is a convergence challenge function. From the closed-form hypothesis, for every $\varepsilon > 0$ there exists N with $d(a_n, \ell) \leq \varepsilon$ for all $n \geq N$. Since $\varepsilon > 0 \implies \varepsilon < 2\varepsilon$, we obtain $d(a_n, \ell) \leq \varepsilon < 2\varepsilon$, hence

$$\exists \ell \in F, \forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1, n \geq N, d(a_n, \ell) < 2\varepsilon. \quad (*)$$

For any $\varepsilon > 0$, applying $(*)$ at $\varepsilon/2 > 0$ gives N such that $d(a_n, \ell) < 2 \cdot (\varepsilon/2) = \varepsilon$ for all $n \geq N$. $\square$

3 Limit of a Sequence

Definition 3.1 (Limit of a Sequence). [def:limit] Let F be an ordered field. We say the sequence $a : \mathbb{N}_1 \rightarrow F$ has a limit of $\ell \in F$ if

$$\forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1, n \geq N, d(a_n, \ell) < \varepsilon.$$

Definition 3.2 (Set of Convergent Sequences). [def:convergent-sequences-set] Let F be an ordered field. The set of convergent sequences in F is

$$\text{Conv}(F) := \{a : \mathbb{N}_1 \rightarrow F \mid a \text{ is convergent}\}.$$

Definition 3.3 (Limit Operator). [def:limit-op] Let F be an ordered field. The limit operator is the function

$$\lim : \text{Conv}(F) \rightarrow F, \quad a \mapsto \text{the unique } \ell \in F \text{ such that } a \text{ has a limit of } \ell.$$

We write

$$\lim a_n = \ell.$$

Remark 3.1 (Well-Definedness of the Limit). [rem:limit-well-defined] The function $\lim$ in definition 3.3 is well-defined for two reasons:

- *Every input has an output:* for every $a \in \text{Conv}(F)$, the definition of convergent sequence guarantees some $\ell \in F$ with a has a limit of ℓ .
- *Every input has exactly one output:* by theorem 1.1, this ℓ is unique.

Theorem 3.1 (Scalar Multiplication of the Limit). [thm:limit-scalar-multiplication] *Let F be an ordered field, let $a \in \text{Conv}(F)$, and let $c \in F$. Then*

$$\lim(c \cdot a_n) = c \cdot \lim a_n.$$

Theorem 3.2 (Addition of the Limit). [thm:limit-addition] *Let F be an ordered field, and let $a, b \in \text{Conv}(F)$. Then*

$$\lim(a_n + b_n) = \lim a_n + \lim b_n.$$

Theorem 3.3 (Linearity of the Limit). [thm:limit-linearity] *Let F be an ordered field, and let $a, b \in \text{Conv}(F)$. Then*

$$\lim(c \cdot a_n + d \cdot b_n) = c \cdot \lim a_n + d \cdot \lim b_n.$$

Theorem 3.4 (Multiplication of the Limit). [thm:limit-multiplication] *Let F be an ordered field, and let $a, b \in \text{Conv}(F)$. Then*

$$\lim(a_n \cdot b_n) = \lim a_n \cdot \lim b_n.$$

Theorem 3.5 (Division of the Limit). [thm:limit-division] *Let F be an ordered field, and let $a, b \in \text{Conv}(F)$ with $\forall n \in \mathbb{N}_1, b_n \neq 0$ and $\lim b_n \neq 0$. Then*

$$\lim \left(\frac{a_n}{b_n} \right) = \frac{\lim a_n}{\lim b_n}.$$

Theorem 3.6 (Order Preservation of Limits). [thm:limit-order-preservation] *Let F be an ordered field, and let $a, b \in \text{Conv}(F)$ with $\forall n \in \mathbb{N}_1, a_n \leq b_n$. Then*

$$\lim a_n \leq \lim b_n.$$

4 Bounded Sequences

Definition 4.1 (Bounded Sequence). [def:bounded-sequence] A sequence $a : \mathbb{N}_1 \rightarrow F$ is said to be *bounded* if

$$\exists M \in F, \forall n \in \mathbb{N}_1, d(a_n, 0) \leq M$$

Theorem 4.1 (Convergent Sequences are Bounded). [thm:convergent-sequences-are-bounded] *If a sequence $a : \mathbb{N}_1 \rightarrow F$ is convergent, it must be bounded. ie,*

$$(\exists \ell \in F, \forall \varepsilon \in F > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1 > N, d(a_n, \ell) < \varepsilon) \implies (\exists M \in F, \forall n \in \mathbb{N}_1, d(a_n, 0) \leq M)$$

Proof. Suppose the convergent sequence a , $\lim a = \ell$.

Take $\varepsilon = 1$, then there exists $N \in \mathbb{N}_1$ such that for all $n \in \mathbb{N}_1$ with $n > N$, $d(a_n, \ell) < 1$.

According to the triangle inequality for distance, and $d(a_n, \ell) < 1$, we have

$$d(0, a_n) \leq d(0, \ell) + d(\ell, a_n) < d(0, \ell) + 1$$

Which means for all $n \in \mathbb{N}_1$ with $n > N$, $d(0, a_n) < d(0, \ell) + 1$.

And for $n \in \mathbb{N}_1$ with $n \leq N$, we have $d(0, a_n) \leq \max\{d(0, a_1), d(0, a_2), \dots, d(0, a_N)\}$.

So, we can take $M = \max\{\max\{d(0, a_1), d(0, a_2), \dots, d(0, a_N)\}, d(0, \ell) + 1\}$. Hence M exists for all convergent sequences a . $\square$

Remark 4.1. Although Convergent $\implies$ Bounded, the converse is not true. A bounded sequence may not be convergent.

For example, the sequence $a_n = (-1)^n$ is bounded (since $d(a_n, 0) = 1$ for all n), but not convergent.

5 Monotone Convergence Theorem

Definition 5.1 (Increasing Sequence). [def:increasing-sequence] A sequence $a : \mathbb{N}_1 \rightarrow F$ is said to be *increasing* if

$$\forall n \in \mathbb{N}_1, a_n \leq a_{n+1}$$

Definition 5.2 (Strictly Increasing Sequence). [def:strictly-increasing-sequence] A sequence $a : \mathbb{N}_1 \rightarrow F$ is said to be *strictly increasing* if

$$\forall n \in \mathbb{N}_1, a_n < a_{n+1}$$

Definition 5.3 (Decreasing Sequence). [def:decreasing-sequence] A sequence $a : \mathbb{N}_1 \rightarrow F$ is said to be *decreasing* if

$$\forall n \in \mathbb{N}_1, a_n \geq a_{n+1}$$

Definition 5.4 (Strictly Decreasing Sequence). [def:strictly-decreasing-sequence] A sequence $a : \mathbb{N}_1 \rightarrow F$ is said to be *strictly decreasing* if

$$\forall n \in \mathbb{N}_1, a_n > a_{n+1}$$

Lemma 5.1 (Strictly Increasing Extends to All Indices). [lem:strictly-increasing-extends] Let $a : \mathbb{N}_1 \rightarrow F$ be strictly increasing. Then $\forall n, m \in \mathbb{N}_1, n > m \implies a_n > a_m$.

Lemma 5.2 (Increasing Extends to All Indices). [lem:increasing-extends] Let $a : \mathbb{N}_1 \rightarrow F$ be increasing. Then $\forall n, m \in \mathbb{N}_1, n \geq m \implies a_n \geq a_m$.

Lemma 5.3 (Strictly Decreasing Extends to All Indices). [lem:strictly-decreasing-extends] Let $a : \mathbb{N}_1 \rightarrow F$ be strictly decreasing. Then $\forall n, m \in \mathbb{N}_1, n > m \implies a_n < a_m$.

Lemma 5.4 (Decreasing Extends to All Indices). [lem:decreasing-extends] Let $a : \mathbb{N}_1 \rightarrow F$ be decreasing. Then $\forall n, m \in \mathbb{N}_1, n \geq m \implies a_n \leq a_m$.

Definition 5.5 (Monotone Sequence). [def:monotone-sequence] A sequence $a : \mathbb{N}_1 \rightarrow F$ is said to be *monotone* if it is either increasing or decreasing.

Theorem 5.1 (Monotone Convergence Theorem). [thm:mct] Let F be an complete ordered field, and let $a : \mathbb{N}_1 \rightarrow F$ be a monotone sequence. If a is bounded, then a is convergent.

Proof. Suppose a is a bounded sequence in F and increasing sequence, which means exist $x = \sup a$ (by the Axiom of Completeness, applied to the nonempty set $\{a_n : n \in \mathbb{N}_1\}$ bounded above in F), and $\forall n, m \in \mathbb{N}_1, n \leq m \implies a_n \leq a_m$.

According to the ε criterion for supremum, $\forall n \in \mathbb{N}_1, a_n \leq x$ and $\forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, a_N > x - \varepsilon$

Adding the increasing property, ie

$$N \leq n \implies a_N \leq a_n \leq x \implies (a_N > x - \varepsilon \implies a_n \geq a_N > x - \varepsilon)$$

We can have

$$\forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1 \geq N, a_n > x - \varepsilon$$

Also,

$$a_n \leq x \wedge \varepsilon > 0 \implies a_n < x + \varepsilon$$

$$a_n > x - \varepsilon \wedge a_n < x + \varepsilon \implies x - \varepsilon < a_n < x + \varepsilon \implies d(a_n, x) < \varepsilon$$

Notice that such x is the supremum of the upper bounded sequence a , it exists.

$$\exists x \in F, \forall \varepsilon \in F, \varepsilon > 0, \exists N \in \mathbb{N}_1, \forall n \in \mathbb{N}_1 \geq N, d(a_n, x) < \varepsilon$$

Which means it is a convergent sequence and its limit is x .

If instead a is decreasing, the same argument applies with $x := \inf a$ (which exists by the Axiom of Completeness, since $\{a_n : n \in \mathbb{N}_1\}$ is nonempty and bounded below), using the dual ε -criterion for the infimum. $\square$

6 Comparison of Sequences

Theorem 6.1 (Comparison for Monotone Sequences). [thm:monotone-comparison-increasing] *Let F be a complete ordered field, and let $(s_n), (t_n) : \mathbb{N}_1 \rightarrow F$ be increasing with $s_n \leq t_n$ for all $n \in \mathbb{N}_1$. If (t_n) is convergent, then*

$$(s_n) \text{ convergent and } \lim s_n \leq \lim t_n.$$

Proof. **TODO** □

Theorem 6.2 (Comparison for Decreasing Sequences). [thm:monotone-comparison-decreasing] *Let F be a complete ordered field, and let $(s_n), (t_n) : \mathbb{N}_1 \rightarrow F$ be decreasing with $s_n \leq t_n$ for all $n \in \mathbb{N}_1$. If (s_n) is convergent, then*

$$(t_n) \text{ convergent and } \lim s_n \leq \lim t_n.$$

Proof. Apply theorem 6.1 to the increasing sequences $(-t_n), (-s_n)$. □

7 Squeeze Theorem

Theorem 7.1 (Squeeze Theorem). [thm:squeeze-theorem] *Let F be an ordered field, and let $a, x, b : \mathbb{N}_1 \rightarrow F$ with $a_n \leq x_n \leq b_n$ for all $n \in \mathbb{N}_1$. If a and b are convergent with $\lim a_n = \lim b_n = L$, then x is convergent and $\lim x_n = L$.*

Proof. **TODO** □